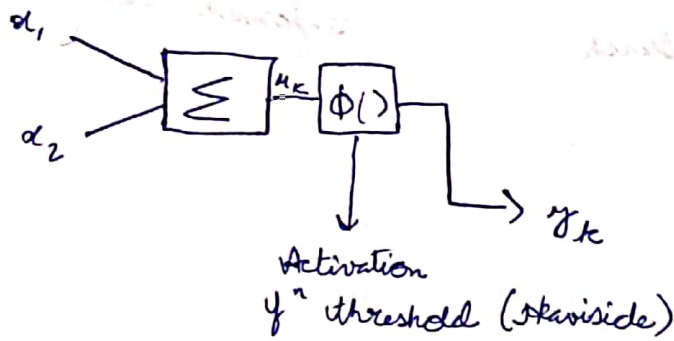


30/8/22

DL

* McCulloch Pitts Model



OR gate

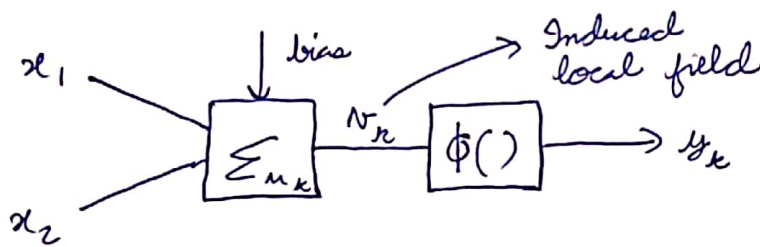
x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	1

$$u_k = x_1 + x_2$$

$$y_k = \phi(u_k)$$

$$= \begin{cases} 1 & u_k \geq 0 \\ 0 & u_k < 0 \end{cases}$$

* McCulloch Pitts Model with Bias



$$u_k = x_1 + x_2$$

$$v_k = u_k + b_k$$

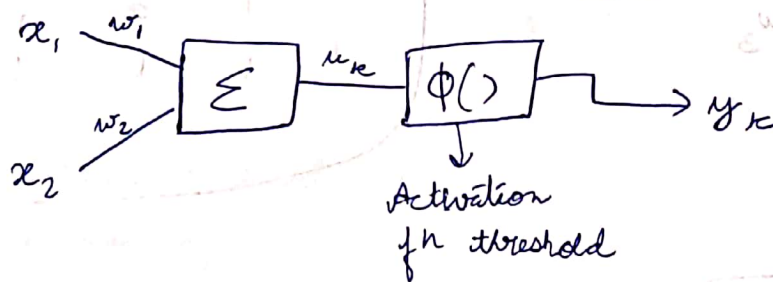
$$y_k = \phi(v_k)$$

$$= \begin{cases} 1 & v_k \geq 0 \\ 0 & v_k < 0 \end{cases}$$

Q, Derive McCulloch-Pitts Model for an OR gate without bias.

Solⁿ:- We can change the threshold f^n . Also, we can take bias. Because for the previous threshold f^n & without bias, y_k will be equal to 1 each time.

* Perceptron Model (~~Structure~~ Features with weights) without bias



$$u_k = w_1 x_1 + w_2 x_2$$

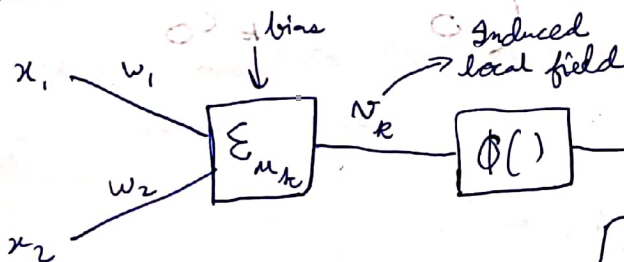
$$y_k = \Phi(u_k)$$

$$= \begin{cases} 1 & u_k \geq 0 \\ 0 & u_k < 0 \end{cases}$$

Note

⇒ We can take any values of w_1 & w_2 . And bias also.

* Perceptron Model with Bias



$$u_k = w_1 x_1 + w_2 x_2$$

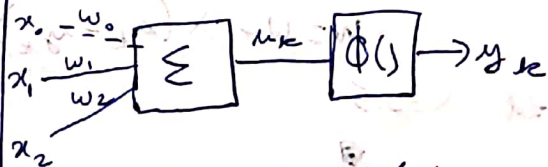
$$v_k = u_k + b_k$$

$$y_k = \Phi(v_k)$$

$$= \begin{cases} 1 & v_k \geq 0 \\ 0 & v_k < 0 \end{cases}$$

Including bias along with I/P

Let, x_0 be bias & w_0 be importance for x_0 .



$$u_k = x_0 w_0 + x_1 w_1 + x_2 w_2$$

$$y_k = \Phi(u_k) = \begin{cases} 1 & u_k \geq 0 \\ 0 & u_k < 0 \end{cases}$$

Q. Design a Perceptron model for OR gate with 3 input values.

Soln: Inputs

x_1, x_2, x_3

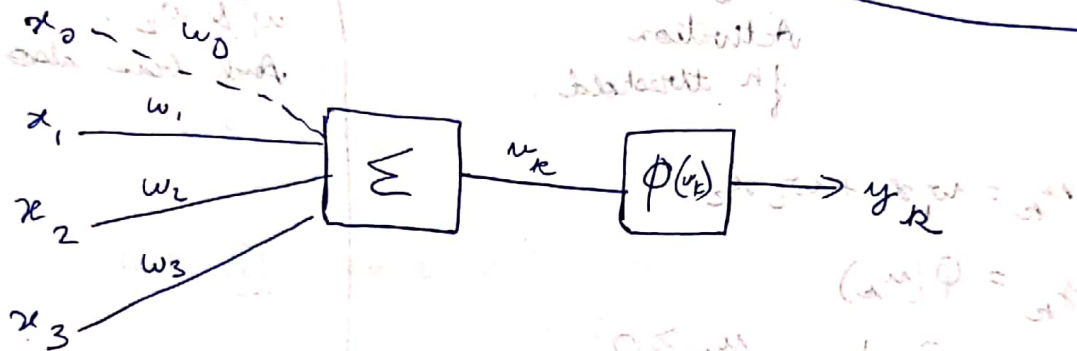
Bias

x_0

Weights

w_0, w_1, w_2, w_3

<u>OR</u>			
x_1	x_2	x_3	y
0	0	0	0
0	0	1	0
0	1	0	0
0	1	1	1
1	0	0	0
1	0	1	1
1	1	0	1
1	1	1	1



$$u_k = \sum_{i=0}^3 x_i w_i$$

$$y_k = \phi(u_k) = \begin{cases} 1 & \text{if } u_k \geq 0 \\ 0 & \text{if } u_k < 0 \end{cases}$$

u_k = actual o/p

y_k = desired o/p

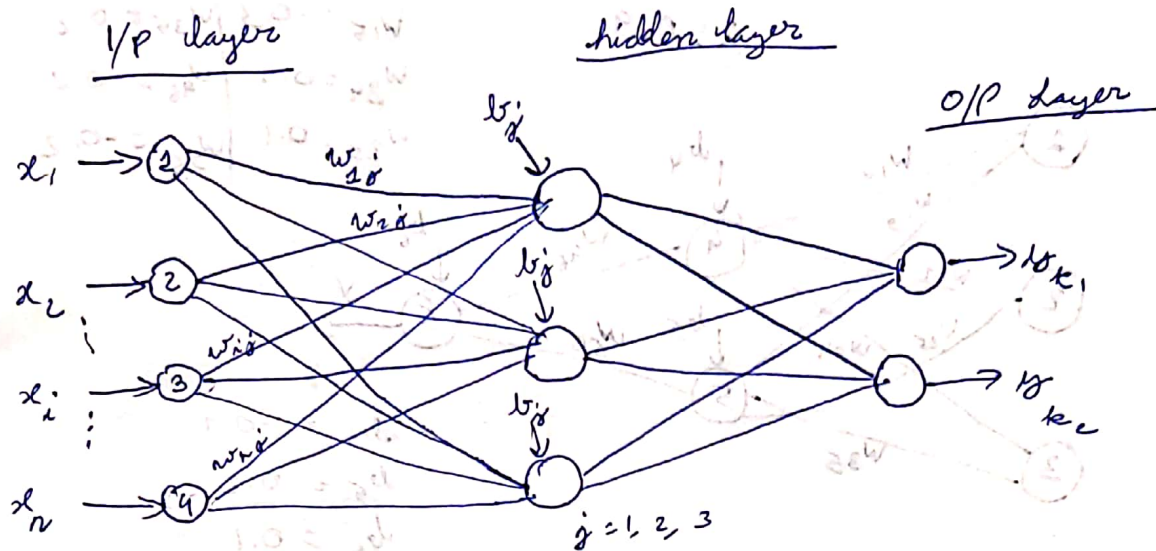
Weight updation

$$W = W + \lambda (u_k - y_k)$$

Let $\lambda = 0.5$

learning rule
 $\lambda \rightarrow$ learning rate
 $0 < \lambda \leq 1$

Multilayer Perceptron



- Fully connected
- Two layer N/W



i) Pre activation Phase ($I_j = \sum_i w_{ij} x_i + b_j$)

ii) Activation Phase ($O_j = \frac{1}{1 + e^{-I_j}}$)

(1) Feed Forward

(2) Back Propagation

Error

O/P $\rightarrow$ hidden layer $\rightarrow$ Input layer (Next Instance)

$$Err_k = O_k(1 - O_k)(T_k - O_k) \quad [\text{error } f^n \text{ of O/P layer}]$$

$$Err_j = O_j(1 - O_j) \sum_k Err_k w_{jk} \quad [\text{hidden layer}]$$

$$w_{ij} = w_{ij} + \lambda Err_j O_i$$

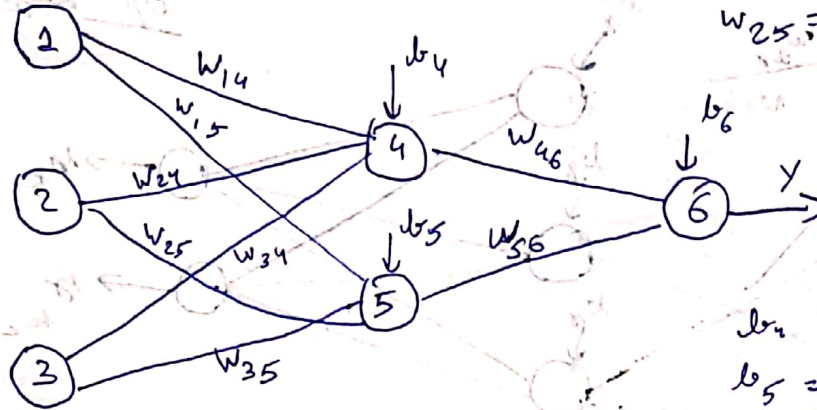
$$b_j = b_j + \lambda Err_j$$

Q. $x = [1, 0, 1]$

$y = 1$

$\lambda = 0.9$

$w_{14} = 0.2$	$w_{34} = -0.5$
$w_{15} = -0.3$	$w_{35} = 0.2$
$w_{24} = 0.4$	$w_{46} = -0.3$
$w_{25} = 0.1$	$w_{56} = -0.2$



$b_4 = -0.4$
 $b_5 = 0.2$
 $b_6 = 0.1$

Use:

$I_1 = 1$	$O_1 = 1$
$I_2 = 0$	$O_2 = 0$
$I_3 = 1$	$O_3 = 1$

$$\begin{aligned}
 I_4 &= w_{14}x_1 + w_{24}x_2 + w_{34}x_3 + b_4 \\
 &= (0.2)(1) + (0.4)(0) + (-0.5)(1) + (-0.4) \\
 &= 0.2 - 0.5 - 0.4 \\
 &= \boxed{-0.7}
 \end{aligned}$$

$$\begin{aligned}
 I_5 &= w_{15}x_1 + w_{25}x_2 + w_{35}x_3 + b_5 \\
 &= (-0.3)(1) + (0.1)(0) + (0.2)(1) + (0.2) \\
 &= -0.3 + 0.2 + 0.2 \\
 &= \boxed{0.1}
 \end{aligned}$$

$$O_4 = \frac{1}{1 + e^{-I_4}} = \frac{1}{1 + e^{-0.7}} = 0.332$$

$$O_5 = \frac{1}{1 + e^{-I_5}} = \frac{1}{1 + e^{-0.1}} = 0.525$$

$$I_6 = W_{46} O_4 + W_{56} O_5 + b_6$$

$$= (-0.3)(0.332) + (-0.2)(0.525) + 0.1$$

$$= -0.0996 - 0.105 + 0.1$$

$$= \boxed{-0.1}$$

$$O_6 = \frac{1}{1 + e^{-I_6}} = \frac{1}{1 + e^{0.1}} =$$

$$= \boxed{0.475}$$

Errors:

$$Err_6 = O_6 (1 - O_6) (T_6 - O_6)$$

$$= \cancel{0.1309} \boxed{0.1309}$$

$$(T_6 = \gamma = 1)$$

$$(O_6 = 0.475)$$

$$Err_5 = O_5 (1 - O_5) W_{56} Err_6$$

$$= \cancel{0.0448} \boxed{-0.0065}$$

$$(O_5 = 0.525)$$

$$W_{56} = -0.2$$

$$Err_4 = O_4 (1 - O_4) W_{46} Err_6$$

$$= \boxed{-0.0087}$$

$$(O_4 = 0.332)$$

$$W_{46} = -0.3$$

$$W_{46} = W_{46} + \lambda Err_6 O_4$$

$$= (-0.3) + (0.9 \times 0.1309 + 0.332)$$

$$= \boxed{-0.2608}$$

$$W_{56} = W_{56} + \lambda Err_6 O_5 = \boxed{-0.1381}$$

$$W_{14} = W_{14} + \lambda Err_4 O_1 = \boxed{0.1921}$$

$$W_{15} = W_{15} + \lambda Err_5 O_1 = \boxed{-0.3058}$$

$$W_{24} = W_{24} + \lambda Err_4 O_2 = \boxed{0.4}$$

$$W_{25} = W_{25} + \lambda Err_5 O_2 = \boxed{0.1}$$

$$W_{34} = W_{34} + \lambda Err_4 O_3 = \boxed{-0.5078}$$

$$W_{35} = W_{35} + \lambda Err_5 O_3 = \boxed{0.1942}$$

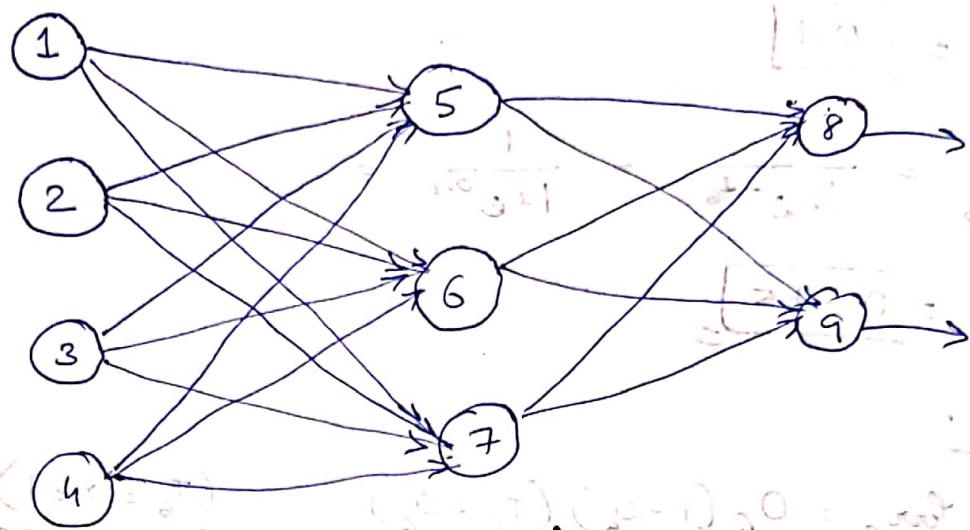
Date
5/9/22

* Feed Forward Neural Network

Input layer

Hidden layer

O/P layer



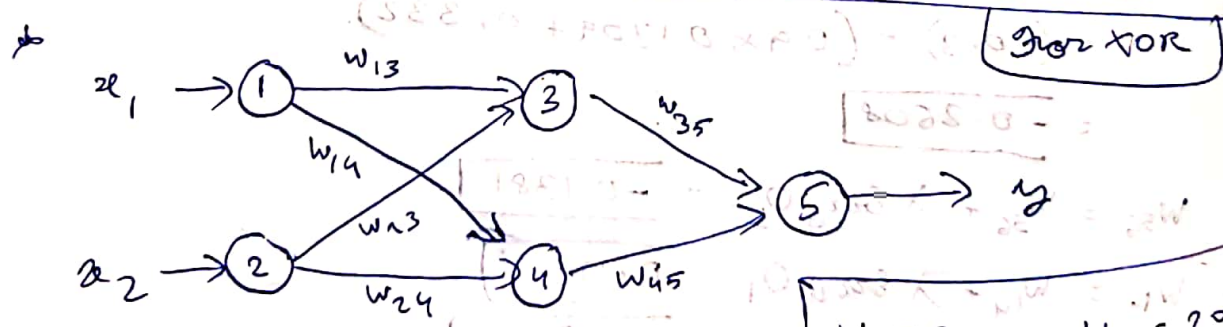
$$I_j = \sum_i w_{ij} x_i + b_j$$

$$O_j = \frac{1}{1 + e^{-I_j}}$$



Deactivation
phase (I)
Activation
phase (0)

XOR problem		
x_1	x_2	y
0	0	0
0	1	1
1	0	1
1	1	0

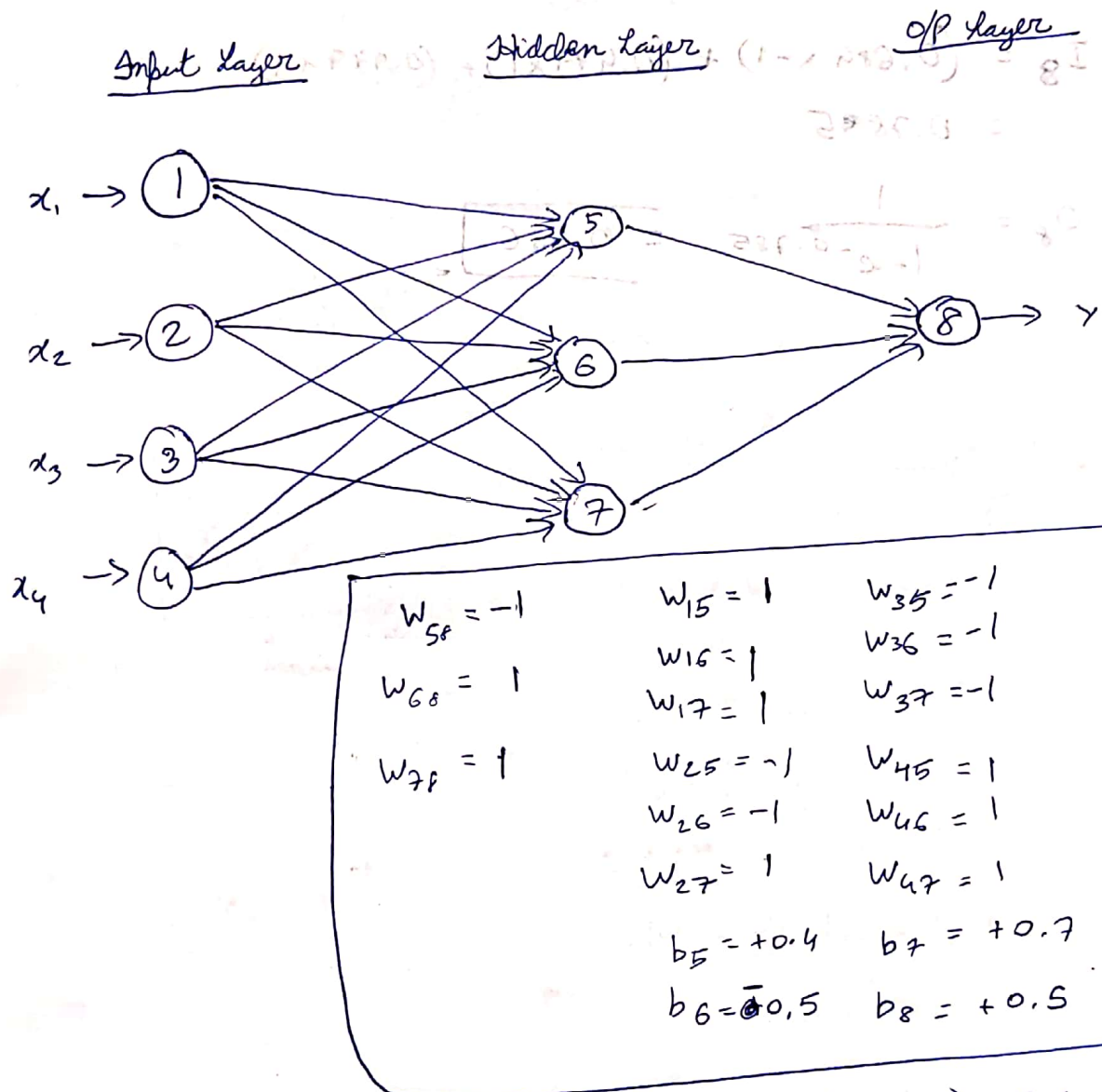


Solⁿ: Pehle $x_1=0, x_2=0$ ke liye
 I_3, I_4, I_5 k O_3, O_4, O_5
nikalo.
Phir $x_1=0$ k $x_2=1$ ke liye
nikalo.

$w_{13} = 20$
 $w_{14} = 20$
 $w_{23} = -20$
 $w_{24} = -20$
 $w_{35} = 20$
 $w_{45} = 20$
 $b_3 = -10$
 $b_4 = 30$
 $b_5 = 30$

x_1	x_2	x_3	x_4	C
5.1	3.5	1.4	0.2	1
4.8	3.0	1.4	0.3	1
7.0	2.2	4.7	1.4	2

solⁿ:- First we need to design the network, we will make (n-1) hidden layers & 1 output layer,



solⁿ:-

$$I_5 = (5.1 \times 1) + (3.5 \times -1) + (1.4 \times -1) + (0.2 \times 1) + 0.4$$

$$= 0.8$$

$$I_6 = -0.1$$

$$I_7 = 8.1$$

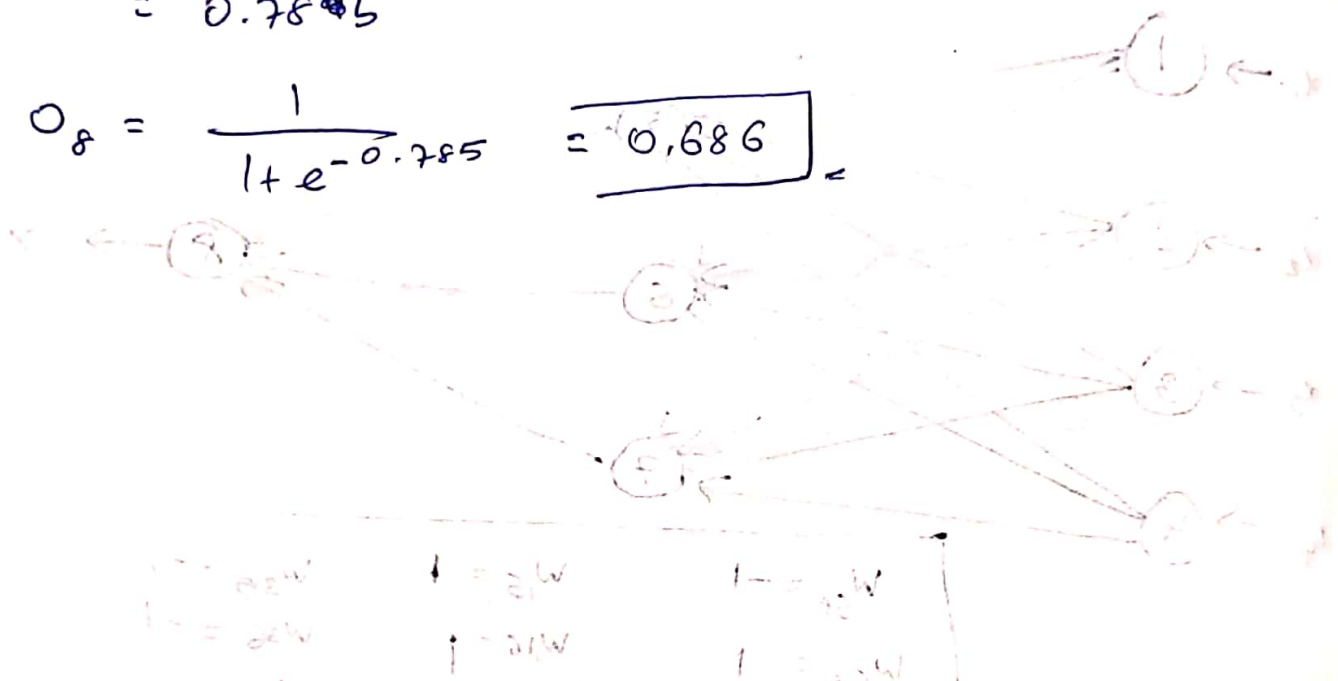
$$O_5 = \frac{1}{1+e^{-I_5}} = \frac{1}{1+e^{-0.8}} = 0.689$$

$$O_6 = \frac{1}{1+e^{-I_6}} = \frac{1}{1+e^{0.1}} = 0.475$$

$$O_7 = \frac{1}{1+e^{-I_7}} = \frac{1}{1+e^{-8.1}} = 0.999$$

$$I_8 = (0.689 \times -1) + (0.479 \times 1) + (0.999 \times 1) = 0.785$$

$$O_8 = \frac{1}{1+e^{-0.785}} = \boxed{0.686}$$



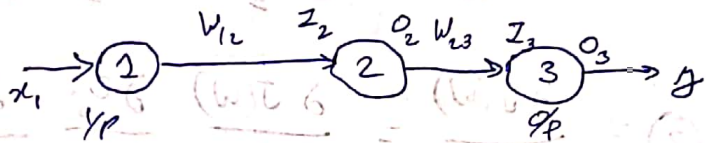
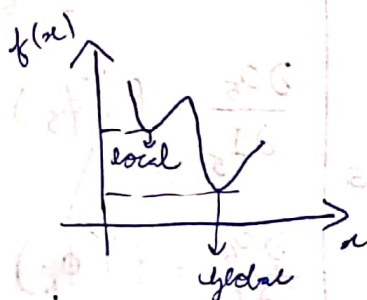
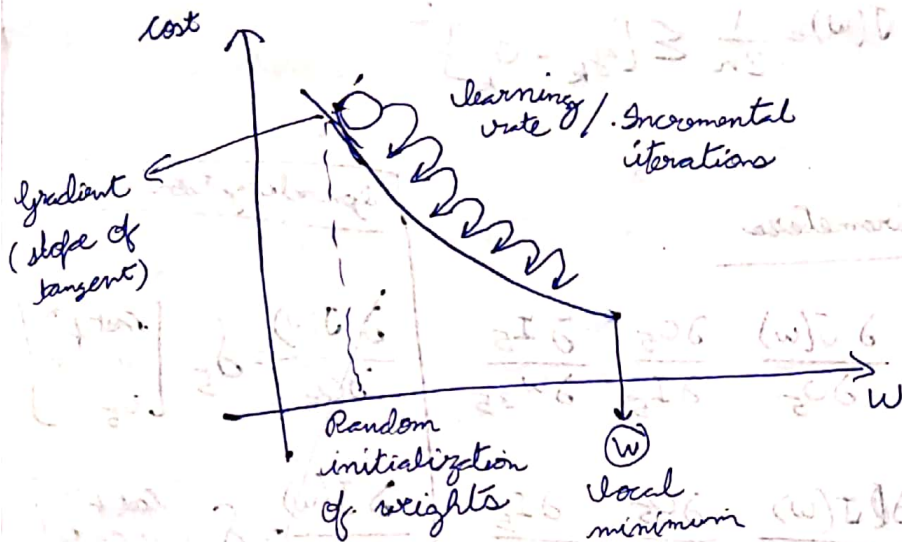
Gradient Descent (Multi Layered Network with Back Propagation)

Date: 7/9/22

* Optimization algorithm - Error minimization

* Train the Neural N/w model.

* To identify local minimum.



$$I_2 = x_1 w_{12}$$

$$O_2 = \phi(I_2)$$

$$I_3 = O_2 w_{23}$$

$$O_3 = \phi(I_3)$$

Cost function:

$$J(w) = \frac{1}{2n} [y_k - o_k]^2$$

y_k - Expected

o_k - Observed

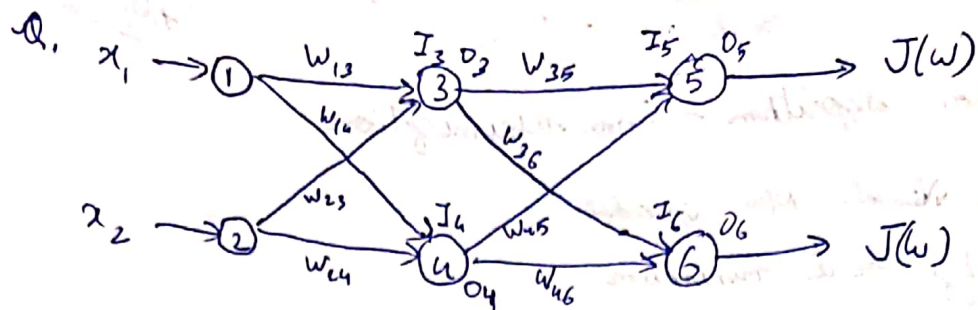
n - No. of instances in a Dataset

Cost fⁿ

$$J(w) = \frac{1}{2n} [y_k - o_k]^2$$

$$\frac{\partial J(w)}{\partial w_{23}} = \frac{\partial J(w)}{\partial O_3} \cdot \frac{\partial O_3}{\partial I_3} \cdot \frac{\partial I_3}{\partial w_{23}}$$

$$\frac{\partial J(w)}{\partial w_{12}} = \frac{\partial J(w)}{\partial O_3} \cdot \frac{\partial O_3}{\partial I_3} \cdot \frac{\partial I_3}{\partial O_2} \cdot \frac{\partial O_2}{\partial I_2} \cdot \frac{\partial I_2}{\partial w_{12}}$$



$$J(w) = \frac{1}{2n} \sum [y_k - o_k]^2$$

Soln:- 2 Layer Parameters

$$\textcircled{1} = \frac{\partial J(w)}{\partial w_{35}} = \frac{\partial J(w)}{\partial o_5} \frac{\partial o_5}{\partial I_5} \frac{\partial I_5}{\partial w_{35}}$$

$$\textcircled{2} = \frac{\partial J(w)}{\partial w_{45}} = \frac{\partial J(w)}{\partial o_5} \frac{\partial o_5}{\partial I_5} \frac{\partial I_5}{\partial w_{45}}$$

$$\textcircled{3} = \frac{\partial J(w)}{\partial w_{36}} = \frac{\partial J(w)}{\partial o_6} \frac{\partial o_6}{\partial I_6} \frac{\partial I_6}{\partial w_{36}}$$

$$\textcircled{4} = \frac{\partial J(w)}{\partial w_{46}} = \frac{\partial J(w)}{\partial o_6} \frac{\partial o_6}{\partial I_6} \frac{\partial I_6}{\partial w_{46}}$$

Therefore from

$$\textcircled{1} = \delta_5 \cancel{f'(\phi_5)} o_3 \quad \checkmark$$

$$\textcircled{2} = \delta_5 \cancel{f'(\phi_5)} o_4 \quad \checkmark$$

$$\textcircled{3} = \delta_6 \cancel{f'(\phi_6)} o_3 \quad \checkmark$$

$$\textcircled{4} = \delta_6 \cancel{f'(\phi_6)} o_4 \quad \checkmark$$

$$\delta_3 = \delta_5 w_{35} f'(\phi_3) + \delta_6 w_{36} f'(\phi_3)$$

$$\delta_4 = \delta_5 w_{45} f'(\phi_4) + \delta_6 w_{46} f'(\phi_4)$$

→ Agla Page

Generalization

$$\frac{\partial J(w)}{\partial o_5} = \delta_5 \left[\text{act. f.} \right]_{o_5}$$

$$\frac{\partial J(w)}{\partial o_6} = \delta_6 \left[\text{act. f.} \right]_{o_6}$$

$$\frac{\partial o_5}{\partial I_5} = f'(\phi_5)$$

$$\frac{\partial o_6}{\partial I_6} = f'(\phi_6)$$

↑
Upper value gets his

$$\frac{\partial J(w)}{\partial o_5} \frac{\partial o_5}{\partial I_5} = \delta_5$$

$$\frac{\partial J(w)}{\partial o_6} \frac{\partial o_6}{\partial I_6} = \delta_6$$

$$\frac{\partial o_5}{\partial I_5} = f'(\phi_5)$$

$$\frac{\partial o_6}{\partial I_6} = f'(\phi_6)$$

$$\textcircled{1} \frac{\partial J(w)}{\partial w_{13}} = \left\{ \frac{\partial J(w)}{\partial o_5} \cdot \frac{\partial o_5}{\partial I_5} \cdot \frac{\partial I_5}{\partial o_3} \cdot \frac{\partial o_3}{\partial I_3} \cdot \frac{\partial I_3}{\partial w_{13}} \right\} +$$

$$\left\{ \frac{\partial J(w)}{\partial o_6} \cdot \frac{\partial o_6}{\partial I_6} \cdot \frac{\partial I_6}{\partial o_3} \cdot \frac{\partial o_3}{\partial I_3} \cdot \frac{\partial I_3}{\partial w_{13}} \right\}$$

$$= \{ \delta_5 f'(\phi_3) w_{35} f'(\phi_3) x_1 \} + \{ \delta_6 f'(\phi_3) w_{36} f'(\phi_3) x_2 \}$$

$$\textcircled{2} \frac{\partial J(w)}{\partial w_{23}} = \left\{ \frac{\partial J(w)}{\partial o_5} \cdot \frac{\partial o_5}{\partial I_5} \cdot \frac{\partial I_5}{\partial o_3} \cdot \frac{\partial o_3}{\partial I_3} \cdot \frac{\partial I_3}{\partial w_{23}} \right\} +$$

$$\left\{ \frac{\partial J(w)}{\partial o_6} \cdot \frac{\partial o_6}{\partial I_6} \cdot \frac{\partial I_6}{\partial o_3} \cdot \frac{\partial o_3}{\partial I_3} \cdot \frac{\partial I_3}{\partial w_{23}} \right\}$$

~~③~~

$$\textcircled{3} \frac{\partial J(w)}{\partial w_{14}} = \left\{ \frac{\partial J(w)}{\partial o_5} \cdot \frac{\partial o_5}{\partial I_5} \cdot \frac{\partial I_5}{\partial o_4} \cdot \frac{\partial o_4}{\partial I_4} \cdot \frac{\partial I_4}{\partial w_{14}} \right\} +$$

$$\left\{ \frac{\partial J(w)}{\partial o_6} \cdot \frac{\partial o_6}{\partial I_6} \cdot \frac{\partial I_6}{\partial o_4} \cdot \frac{\partial o_4}{\partial I_4} \cdot \frac{\partial I_4}{\partial w_{14}} \right\}$$

$$\textcircled{4} \frac{\partial J(w)}{\partial w_{24}} = \left\{ \frac{\partial J(w)}{\partial o_5} \cdot \frac{\partial o_5}{\partial I_5} \cdot \frac{\partial I_5}{\partial o_4} \cdot \frac{\partial o_4}{\partial I_4} \cdot \frac{\partial I_4}{\partial w_{24}} \right\} +$$

$$\left\{ \frac{\partial J(w)}{\partial o_6} \cdot \frac{\partial o_6}{\partial I_6} \cdot \frac{\partial I_6}{\partial o_4} \cdot \frac{\partial o_4}{\partial I_4} \cdot \frac{\partial I_4}{\partial w_{24}} \right\}$$

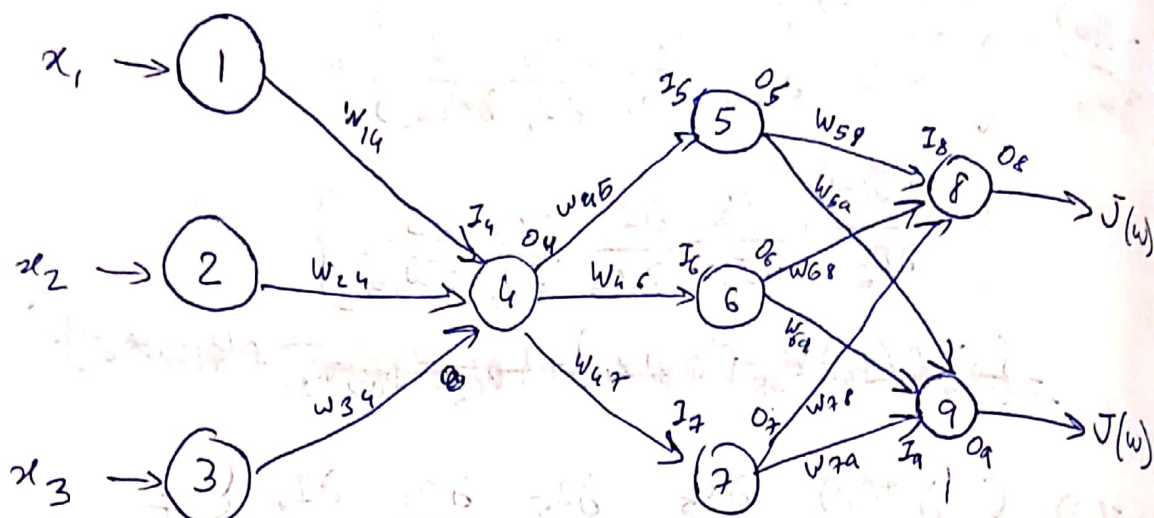
$$\textcircled{1} \equiv \{ \delta_5 w_{35} f'(\phi_3) x_1 \} + \{ \delta_6 w_{36} f'(\phi_3) x_2 \} \equiv x_1 \delta_3$$

$$\textcircled{2} \equiv \{ \delta_5 w_{35} f'(\phi_3) x_2 \} + \{ \delta_6 w_{36} f'(\phi_3) x_2 \} \equiv x_2 \delta_3$$

$$\textcircled{3} \equiv \{ \delta_5 w_{45} f'(\phi_4) x_1 \} + \{ \delta_6 w_{46} f'(\phi_4) x_1 \} \equiv x_1 \delta_4$$

$$\textcircled{4} \equiv \{ \delta_5 w_{45} f'(\phi_4) x_2 \} + \{ \delta_6 w_{46} f'(\phi_4) x_2 \} \equiv x_2 \delta_4$$

Q.



$$w_{59} = \frac{\partial J(w)}{\partial w_{59}} = \frac{\partial J(w)}{\partial o_8} \frac{\partial o_8}{\partial I_8} \frac{\partial I_8}{\partial w_{59}}$$

$$\frac{\partial J(w)}{\partial w_{58}} =$$

$$\frac{\partial J(w)}{\partial w_{68}} =$$

$$\frac{\partial J(w)}{\partial w_{69}} =$$

$$\frac{\partial J(w)}{\partial w_{78}} =$$

$$\frac{\partial J(w)}{\partial w_{79}} =$$

Date
13/9/22

Empirical Risk ~~Management~~ Minimization

Date
14/9/22

* Radial Basis Function

* Radial Basis Neural Network

- I/P layer, hidden layer, O/P layer.

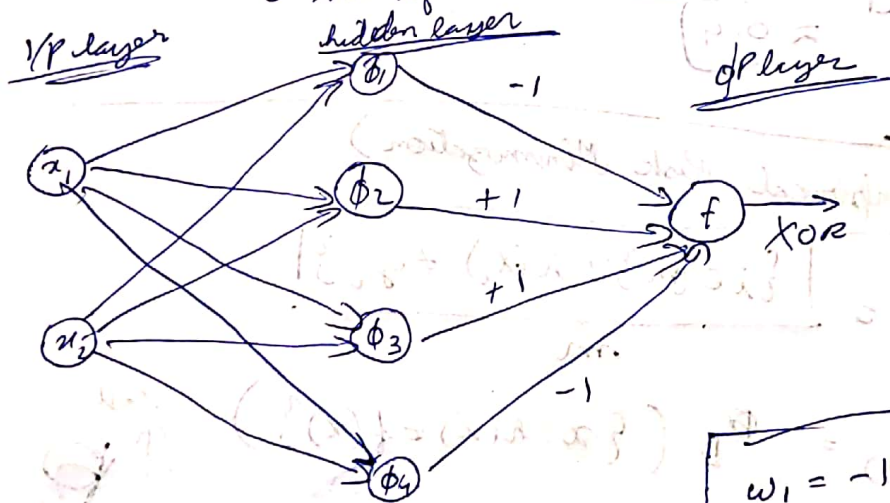
- one hidden layer.

$$\cdot h(x) = e^{-\left(\frac{(x-c)^2}{r^2}\right)}$$

$$\cdot f(x) = \sum_{j=1}^m w_j h_j(x)$$

$$\cdot m > n$$

m = no. of nodes in hidden layer
= no. of nodes in I/P layer.



c

$\phi_1 \Rightarrow (0,0)$
 $\phi_2 \Rightarrow (0,1)$
 $\phi_3 \Rightarrow (1,0)$
 $\phi_4 \Rightarrow (1,1)$

$$r = \sqrt{2}$$

$$\begin{aligned} w_1 &= -1 \\ w_2 &= +1 \\ w_3 &= +1 \\ w_4 &= -1 \end{aligned}$$

$$RBF(x, c)$$

$$I = |x - c|$$

$$= \sqrt{(x_1 - c_1)^2 + (x_2 - c_2)^2}$$

$$I = \sum w_i x_i$$

$$o = f(I)$$

O/P layer

XOR

$$\begin{aligned} w_1 &= -1 \\ w_2 &= +1 \\ w_3 &= +1 \\ w_4 &= -1 \end{aligned}$$

y, p	ϕ_1	ϕ_2	ϕ_3	ϕ_4	$\sum w_i \phi_i$	$\%r$
0 0	1	0.6	0.6	0.4	-0.2	0
0 1					0.2	1
1 0					-0.2	1
1 1					0.2	0

$$\phi_1 = \frac{e^{-|x - c_1|^2}}{e^{-|x - c_1|^2} + e^{-|x - c_2|^2}} = \frac{e^{-|(0-0)+(0-0)|}}{e^{-|x - c_1|^2} + e^{-|x - c_2|^2}} = e^{-0} = 1$$

$$\phi_2 = e^{-1/2} = 0.6$$

$$\phi_3 = e^{-1/2} = 0.6$$

$$\phi_4 = e^{-1} = 0.4$$

ERM (Empirical Risk Minimization)

$$L_{S(n)} = \frac{|\{i \in \{n\} : h(x_i) \neq y_i\}|}{n}$$

$$L_{D, f(h)} = D(\{x : h(x) \neq f(x)\})$$

$$ERM = \argmin_{h \in H} L_{S(n)}, h \in H$$

